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“Project: SecureMail System”

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Summary of Applied Corrections This revised submission addresses all four (4) technical gaps identified in the grader's feedback. Each correction is documented below with before/after comparisons and updated diagrams.

#	Issue	Correction applied	Affected Section
1	DFD Hierarchy Violation	Renumbered child processes of Process 2.0 from 1.1 → 2.1, ensuring parent-child traceability	Phase 4.1 – Level 2 DFD
2	ERD Text Labels	Replaced "1-1" / "1-many" text with standard Crow's Foot graphical symbols (lines, circles, feet)	Phase 4.2 – ERD
3	Ambiguous NFRs	Replaced all "real-time" language with specific measurable time-bound thresholds (ms/seconds)	SRS – Non-Functional Requirements
4	Use Case Symbols	<<include>> and <<extend>> changed from solid lines to dashed arrows per UML 2.x standard	Use Case Diagram

Phase 4.0 - DFD Overview Level 0 and Level 1

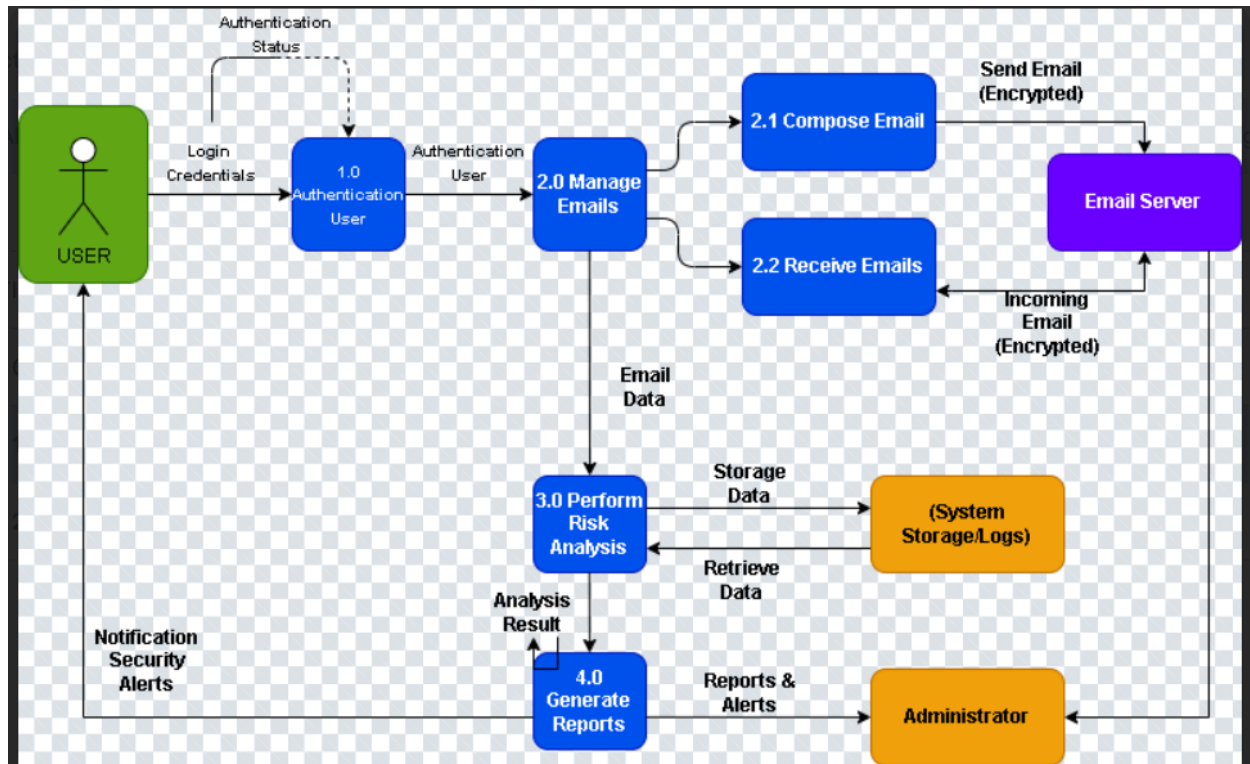
Level 0 DFD - Context Diagram

The Level 0 DFD (Context Diagram) represents the entire SecurMail System as a single process. It shows all external entities that interact with the system and the high level data flows between them, without revealing internal processes.



Level 1 DFD - Major Processes

The Level 1 DFD decomposes the context diagram into three major processes. Each process interacts with external entities and shared data stores. Process 1.0 handles user authentication, Process 2.0 performs risk analysis on login events, and Process 3.0 manages email composition and delivery. Data stores include D1: User Database, D2: Login Log, and D3: Message Store.



1. DFD Hierarchy Correction (Phase 4.1)

Problem

The Level 2 Detailed Decomposition had a critical hierarchical error. The first sub-process of Process 2.0 (Perform Risk Analysis) was labeled 1.1, which structurally implies it belongs to Process 1.0 (Authenticate User). In a properly numbered DFD, the children of Process N.0 must always carry the prefix N.x.

DFD Label	2.0 children: 2.1, 2.2, 2.3, 2.4
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Rule



- Process 1.0 → children: 1.1, 1.2, 1.3 ...
- Process 2.0 → children: 2.1, 2.2, 2.3 ... ← was broken; now fixed
- Process 3.0 → children: 3.1, 3.2, 3.3 ...

Level 2 DFD – SecureMail System

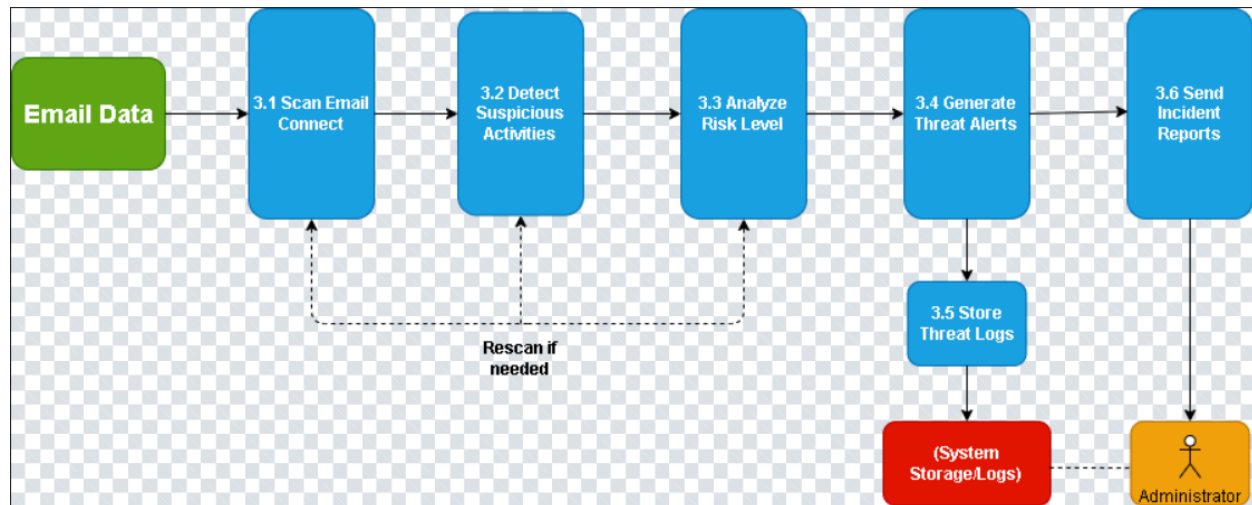


Figure 1: Level 2 DFD — sub-processes of Process 2.0 now correctly labeled 2.1 and 2.2

2. ERD Crow's Foot Notation (Phase 4.2)

Problem

The ERD used plain text labels such as "1-1" and "1-many" instead of the mandatory Crow's Foot graphical symbols. Text labels fail to communicate whether a relationship is mandatory (one or more) or optional (zero or more), which a database architect needs to correctly design the physical schema.



ERD Notation	Crow's Foot: o, , >< graphical symbols
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Crow's Foot Symbol Reference

Symbol Name	Meaning	SecureMail Example
Double bar //	Exactly One (mandatory)	USER must have exactly one user_id
Single bar	One or More (mandatory many)	USER sends one or more MESSAGES
Crow's Foot >	Zero or More (optional many)	MESSAGE may have zero or more ATTACHMENTS
Circle o	Zero (optional / nullable)	LOGIN_LOG — optional risk_score field

ERD – SecureMail System

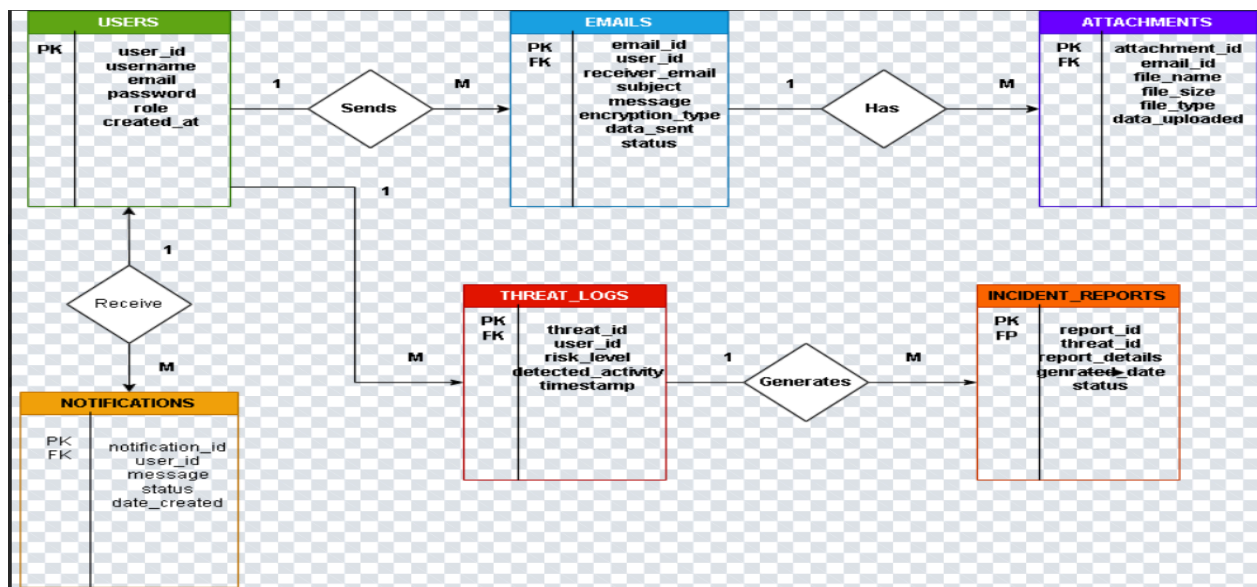


Figure 2: ERD — Crow's Foot graphical symbols replace all text cardinality labels



3. Non-Functional Requirements — Measurable Thresholds

Problem

"Real-time" is a marketing term, not a technical requirement. It is neither testable nor implementable because it provides no threshold for the development or QA team to validate against. Per IEEE 830 SRS standards, every NFR must be specific, measurable, and verifiable.

NFR Language	Specific time-bound thresholds in milliseconds/seconds
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NFR Table

NFR ID	Measurable	Rationale
NFR-01	The system shall calculate a risk score and trigger an alert within 500 ms of a login attempt under normal load.	Provides a testable, measurable threshold per IEEE 830 SRS standards
NFR-02	The system shall encrypt and deliver a composed email to the recipient's inbox within 2 seconds for messages ≤ 1 MB	Removes ambiguity; allows QA team to write objective performance tests.
NFR-03	The system shall load and render a user's inbox (up to 50 messages) within 1.5 seconds at 95th percentile response time.	Percentile-based metric accounts for real-world variability.
NFR-04	All client-server communication shall use TLS 1.2 or higher. Any request not using HTTPS shall be rejected with HTTP 403.	Specifies protocol version; eliminates vague "secure" language.

Why This Matters

- A QA engineer cannot write a test for "real-time" — they can write one for " ≤ 500 ms".
- SLAs with clients require measurable commitments, not qualitative promises.
- Developers need thresholds to make informed architecture decisions (caching, indexing, queuing).



4. Use Case Diagram — <> / <> Line Style

Problem

UML 2.x specifies that <> and <> stereotyped relationships MUST be shown as dashed arrows (open arrowhead, dashed line). Solid lines are reserved for base associations (actor ↔ use case). Using solid lines for <>/<> makes it impossible to distinguish a base actor-association from a structural relationship between use cases.

Use Case Lines	Dashed open-arrowhead lines per UML 2.x standard
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NFR Table

Relationship Type	Line Style	SecureMail Example
Association	Solid line, no arrowhead	User — Login to System
<<include>>	Dashed line, open arrowhead	Login - - - > Authenticate User
<<extend>>	Dashed line, open arrowhead	Send Email - - - > Flag Suspicious Activity
Generalization	Solid line, closed arrowhead	Admin —> User

Use Case Diagram – SecureMail System

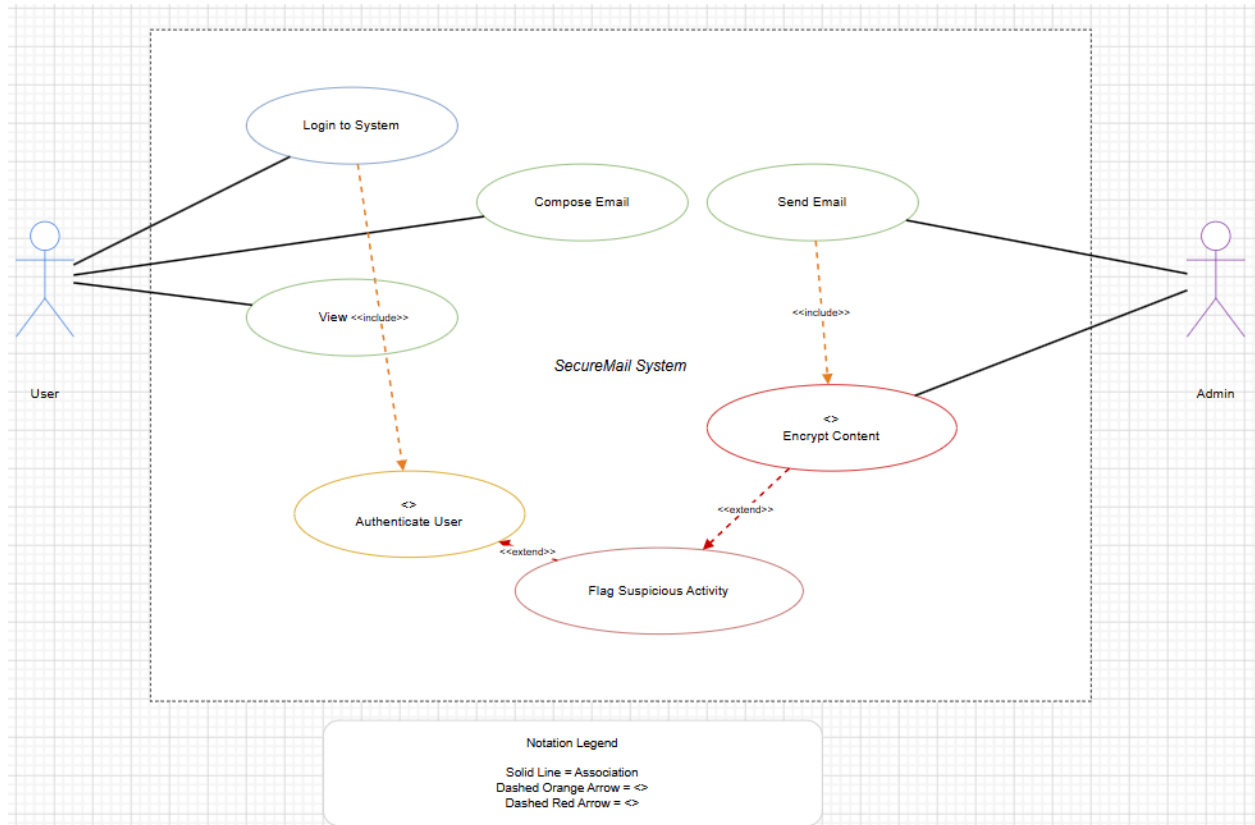


Figure 3: Corrected Use Case Diagram — dashed arrows for << >> and << >> per UML 2.x

Closing Statement

All four corrections from the grader's technical feedback have been applied to this revised submission of the SecureMail System project. The changes made are:

- **DFD Hierarchy** — all child processes now carry the correct parent prefix (1.x, 2.x, 3.x) ensuring traceable data flows throughout the Level 2 decomposition.
- **ERD Notation** — all cardinality labels have been replaced with standard Crow's Foot graphical symbols (bars, circles, feet), allowing a database architect to determine mandatory vs. optional relationships at a glance.
- **"Real-Time" NFRs** — every ambiguous performance requirement has been replaced with a specific, measurable, time-bound threshold (e.g., ≤ 500 ms, ≤ 2 seconds) per IEEE 830 SRS standards.
- **Use Case Symbols** — all << >> and << >> relationships now use dashed openarrowhead lines per UML 2.x specification; solid lines are reserved for actor associations only.